CS131 FINAL PROJECT • DEMO DAY

What Actually Works at 10 m?

An honest study of static, texture, and blade-motion cues for wind-turbine detection in Sentinel-2

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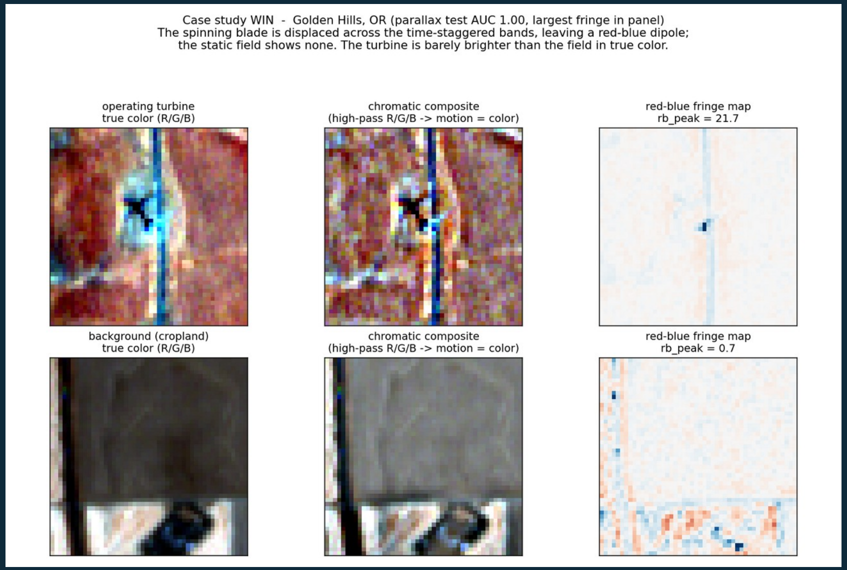


Operating turbine (top) vs background (bottom): the blade-motion fringe is the signal.

MOTIVATION & SIGNIFICANCE

Why detect turbines that are actually turning?

Free, every 5 days

Sentinel-2 is free and global. Official inventories (USWTDB, EIA-860) lag by months; sub-meter aerial imagery is flown only every 2-3 years and cannot nowcast.

Rotation = generation

The cue fires only on spinning blades, a direct proxy for whether a turbine is generating, not just installed. Idle turbines are correctly invisible.

Data-center power diligence

For investors in data centers, pairing which nearby turbines are spinning with their capacity estimates the real wind share feeding a project's grid region, and tests its clean-power claims.

The challenge: at 10 m a turbine is only a few pixels, and on bright or textured ground it is resolved but not locally salient, so appearance-based detection fails on most scenes.

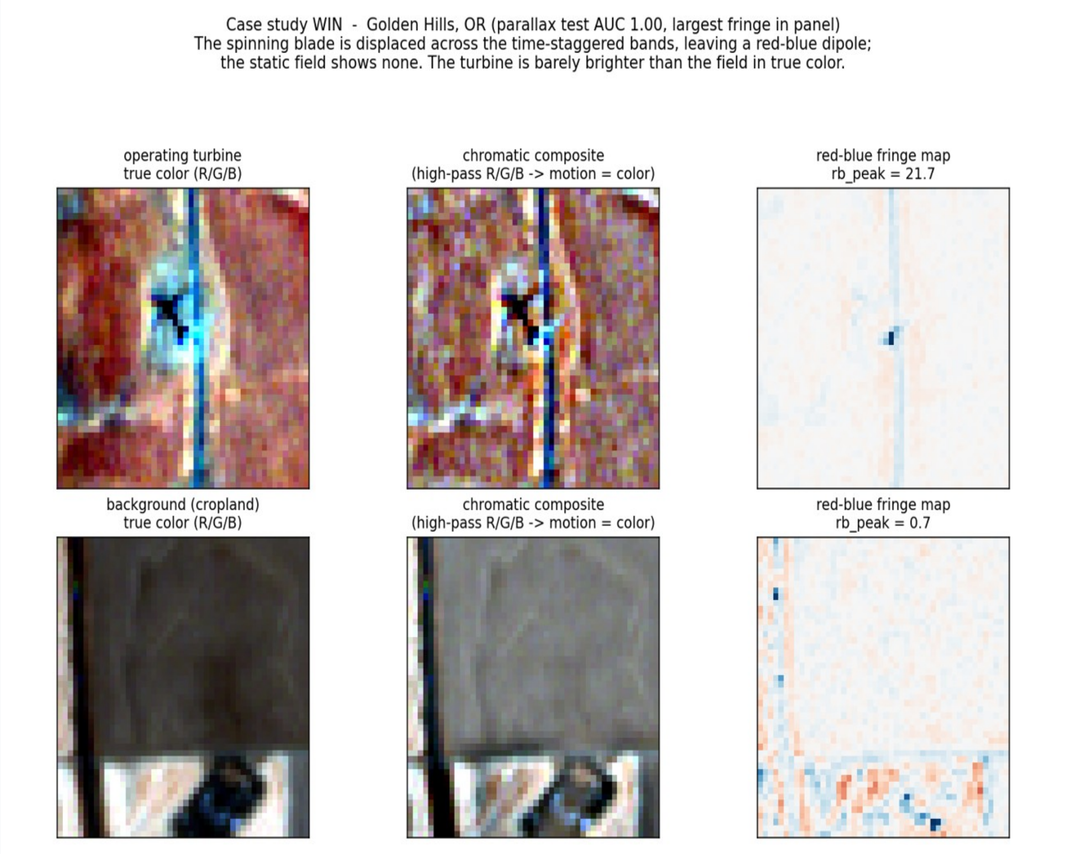
THE KEY IDEA

Read blade motion from a sensor artifact

Sentinel-2 exposes its color bands at slightly different times: **green +0.324 s and red +1.005 s after blue**, because the detectors are offset on the focal plane.

A blade tip moves tens of meters in that window, so an operating turbine sits at displaced pixels across the bands and leaves a localized **red-blue color fringe**. A static object (field, building, pad) leaves none. We engineer 19 interpretable features from that fringe, from scratch.

This is the exact signal the leading 10 m detector (Mandroux et al.) deliberately discards by using the blue band only.



Real Sentinel-2, Golden Hills OR: the turbine is barely brighter in true color, yet a sharp red-blue dipole appears.

METHODOLOGY

Four cue families, one fair benchmark

CS131 static primitives

Gaussian, Sobel, Hough, Harris, matched template. From scratch. The documented fallback.

Static salience

Center-vs-surround brightness z-scores. The binding appearance cue at 10 m.

HOG + linear SVM

A standard generic-appearance baseline (96-D descriptor).

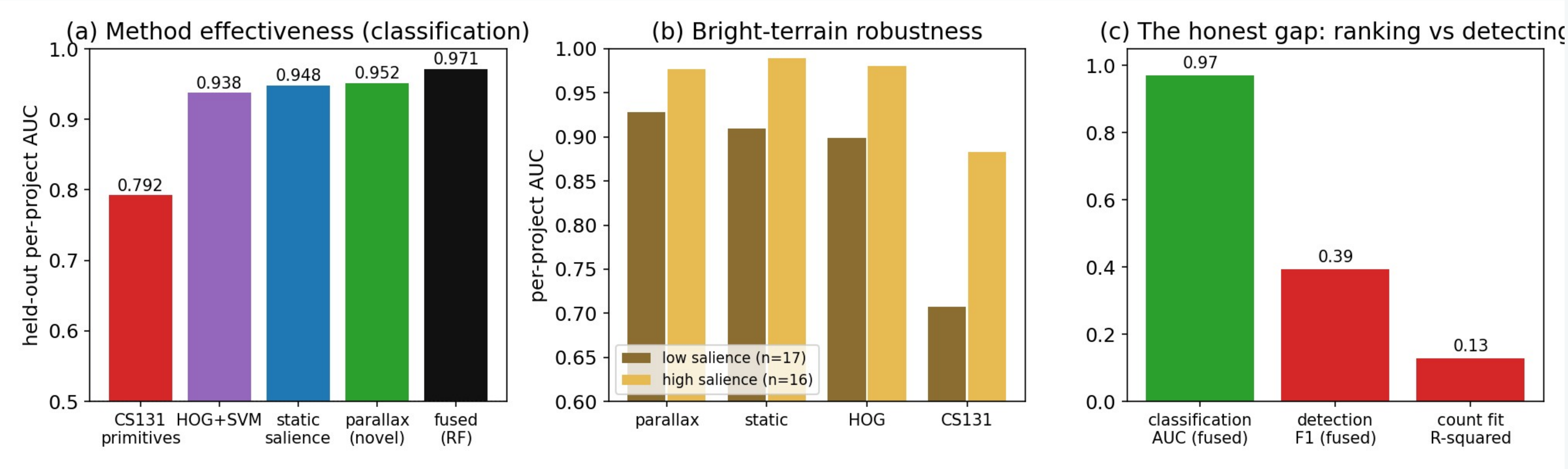
Band-parallax (novel)

19 interpretable blade-motion features from the inter-band fringe.

Benchmark: 173 US projects, 35 states · 10,634 patches · grouped by whole farm (no leakage) · GroupKFold CV · one locked test

RESULTS

What works: motion beats appearance at 10 m



Parallax alone 0.952

best single family, above HOG (0.938) and salience (0.948)

Fused 0.971

parallax + salience; adding HOG or CS131 does not help

Robust on bright terrain

0.928 where appearance collapses (CS131 falls to 0.707)

Real motion, and where it wins or fails

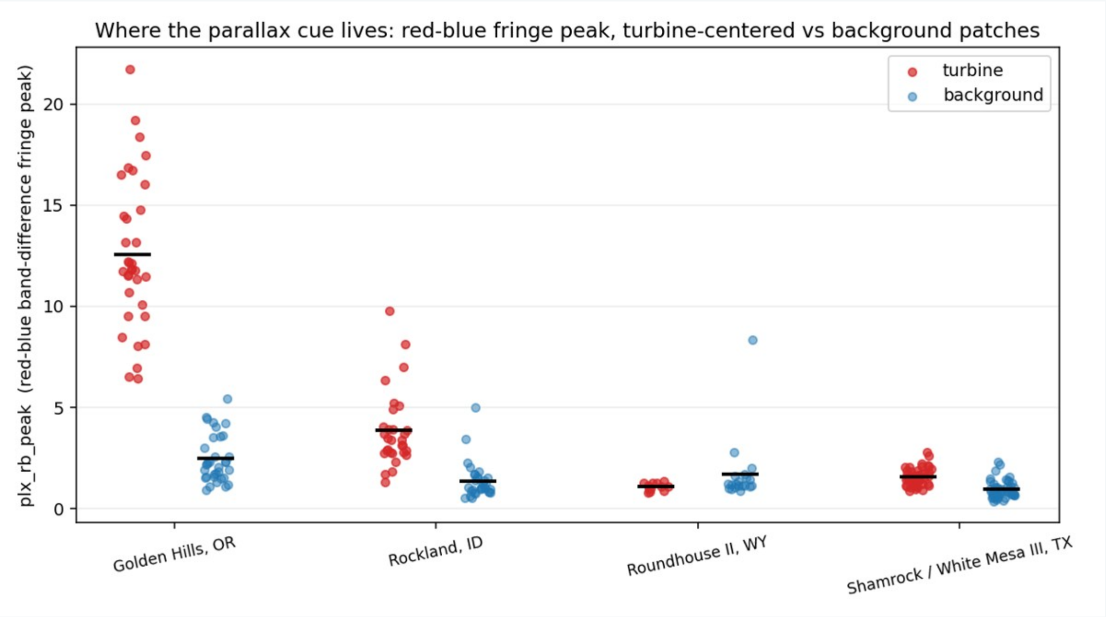
Is it motion or just contrast?

- Residual after removing contrast: AUC 0.763
- Within fixed-contrast bins: parallax 0.759 vs salience 0.527 (chance)
- Label-shuffle control: 0.490 (no leakage)

The cue measures rotation, not presence.

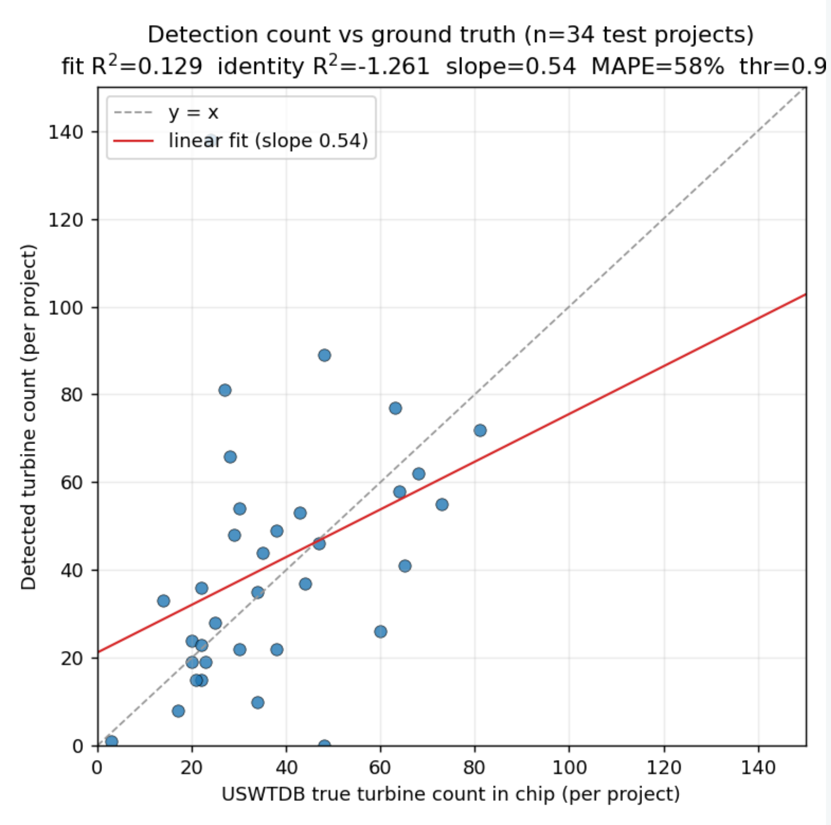
Wins on spinning fleets: Golden Hills, Rockland (even on hard terrain).

Fails on idle fleets: Roundhouse and Shamrock (48 visible turbines, 0 detected) were not turning, so there is no fringe to read.



Red-blue fringe peak, turbine vs background, on four re-fetched projects.

The honest gap: ranking is not detecting



0.97
classification AUC (rank a candidate point)

0.39
object F1 as a real detector

0.13
count R-squared vs ground truth

Learnings

- Ranking pre-placed points is far easier than committing to detections.
- Random-background negatives make the AUC optimistic.
- Idle turbines cap recall against installed-count truth.

TAKEAWAYS

An honest accounting of what each cue is worth

Motion is a real cue

Free Sentinel-2 inter-band parallax is an interpretable, contrast-separable signal of an operating turbine, and the most salience-robust family on bright terrain.

Textbook primitives mostly fail

At 10 m the CS131 family reduces to one Sobel-energy feature; Hough is at chance and HOG adds nothing over the engineered cues.

Be honest about the gap

As a real detector the result is weak (F1 0.39) and idle turbines are invisible. This is a rigorous proof-of-concept, not a counting product.

Future: hard negatives · count benchmark vs EIA-860 · multi-date 5-day nowcast